

JAMES RUSE AGRICULTURAL HIGH SCHOOL
YEAR 11 MATHEMATICS (2 UNIT)
PRELIMINARY EXAMINATION 2005

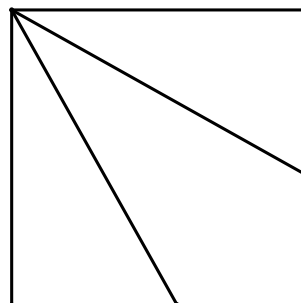
QUESTION 1	Marks
(a) Solve for k : $(4^2)^{3k} = 1$	1
(b) Simplify $\log_4 128$	2
(c) Factorise and simplify: $\frac{x^3 - 27}{x^2 - 9}$	2
(d) Solve for x : $\frac{3x-5}{2} - \frac{9x-1}{5} < -6$	2
(e) Simplify $\frac{\log_7 8}{\log_7 2}$	2
(f) Rationalize the denominator: $\frac{4\sqrt{2}-3}{5+\sqrt{2}}$	2
(g) Simplify $\sqrt{(x+1)^2}$	1
(h) Solve for x : $ 5x-2 = 6x-12$	2
(i) Simplify: $\tan(90^\circ + \theta)$	1

QUESTION 2 (START A NEW PAGE)	Marks
(a) Differentiate with respect to x :	
(i) $9x - x^2\sqrt{x}$	2
(ii) $(x^2 - 1)^{30}$	2
(iii) $\frac{4x-9}{x^3}$	2
(iv) $e^{2x}(e^{4x} - 1)$	2
(v) $x(x+2)^6$	2
(vi) $\sin^2 x$	1
(b) Show that $\frac{d}{dx} \left(\frac{10}{\sqrt{x+3} - \sqrt{x-2}} \right) = \frac{1}{\sqrt{x+3}} + \frac{1}{\sqrt{x-2}}$	2
(c) Show that $f(x)$ is a monotonic decreasing function if $f(x) = \frac{1}{(x+2)^4}$ for $x > -2$.	2

QUESTION 3 (START A NEW PAGE)

Marks

- (a) Find the equation of the normal to the curve $y = 4 \ln(x^2 - 4)$ at the point where $x = 3$. 3
- (b) Find an expression without logarithms if: $2 \log_2(xy) - 3 \log_2(x + y) = 4$. 2
- (c) If α and β are the roots of $4x^2 - 5x + 7 = 0$, find the value of $\alpha^2 + \beta^2$. 2
- (d) Shade the region defined by: $(x + 3)^2 + y^2 \leq 9$. 1
- (e) ABCD is a square with $CX = CY$.
Prove that $\angle AXC = \angle AYC$. 4



- (f) Find the coordinates of the inflexion point on the curve $P(x) = x^3 - 6x^2 + 11x - 6$. 3
Fully justify your answer.

QUESTION 4 (START A NEW PAGE)

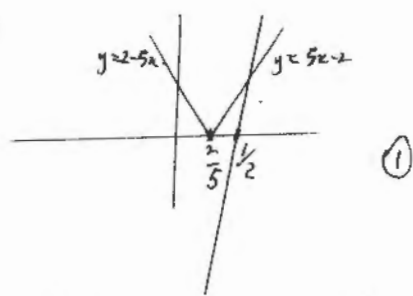
Marks

- (a) Show that quadrilateral ABCD is a parallelogram if its vertices are the points: $A(-14,18)$, $B(-8,27)$, $C(8,12)$ and $D(2,3)$. 3
- (b) Find the value of x , correct to the nearest minute in the domain $-90^\circ \leq x \leq 90^\circ$, if $2 \sin x + 3 \cos x = 0$. 2
- (c) The area A of an irregular shape is given by: $A = 150 - 32y + 2y^2$ where $y = 5 + 4x - x^2$ and x is the distance measured from a fixed point O . 4
- (i) Show that when $x = 0.5$, $\frac{dA}{dx} = -15$. 2
- (ii) Find the value of $\frac{dA}{dx}$ when $x = 1.5$ and $x = 2.5$. 2
- (iii) Find the values of x for the relative minimum and relative maximum values of the area A . 4
- (iv) Graph the area function A versus x in the domain $0 \leq x \leq 4$. 2

2

2

(d)



(1)

$\therefore 6x - 12 = 5x - 2$
 $x = 10 \text{ only.}$

(1)

(i) $\tan(90^\circ + \theta) = \tan[90^\circ - (-\theta)]$
 $= \cot(-\theta)$
 $= -\cot \theta$

(1)

(ii) $\frac{d}{dx} 9x - x^2 \sqrt{x} = \frac{d}{dx} (9x - x^{5/2})$
 $= 9 - \frac{5}{2} x \sqrt{x}$

(2)

(iii) $\frac{d}{dx} (x^2 - 1)^{30} = 60x (x^2 - 1)^{29}$

(2)

(iv) $\frac{d}{dx} \frac{4x - 9}{x^3} = \frac{d}{dx} 4x^{-2} - 9x^{-3}$
 $= -\frac{8}{x^3} + \frac{27}{x^4}$

(2)

(v) $\frac{d}{dx} e^{2x} (e^{4x} - 1) = \frac{d}{dx} e^{6x} - e^{2x}$
 $= 6e^{6x} - 2e^{2x}$

(2)

(vi) $\frac{d}{dx} x(x+2)^6 = (x+2)^6 + 6x(x+2)^5$
 OR $(x+2)^5 (7x+2)$

(2)

(vii) $\frac{d}{dx} [\sin x]^2 = 2 \sin x \cos x$

(1)

2(b) $\frac{d}{dx} \frac{10}{\sqrt{x+3} - \sqrt{x-2}}$
 $= \frac{d}{dx} 10 [\sqrt{x+3} + \sqrt{x-2}]$
 $= \frac{d}{dx} \frac{10}{3+2} [\sqrt{x+3} + \sqrt{x-2}]$

(1)

$= 2 \left[\frac{1}{2\sqrt{x+3}} + \frac{1}{2\sqrt{x-2}} \right]$
 $= \frac{1}{\sqrt{x+3}} + \frac{1}{\sqrt{x-2}}$

(1)

(c) $\frac{d}{dx} \frac{1}{(x+2)^4} = \frac{-4}{(x+2)^5}$

(1)

i.e. $f'(x) < 0$ as $x+2 > 0$
 for all x
 $\therefore$ monotonic decreasing $x > -2$

3(a) $y = 4 \ln(x^2 - 4)$
 $\frac{dy}{dx} = \frac{8x}{x^2 - 4}$

(1)

At $x=3$ $m = \frac{24}{5}$

(1)

$m_{\text{normal}} = -\frac{5}{24}$

(1)

$\therefore x=3$ $y = 4 \ln 5$
 Eqn normal.

$y - y_1 = m(x - x_1)$
 $y - 4 \ln 5 = -\frac{5}{24}(x - 3)$

(1)

$5x + 24y - 96 \ln 5 - 15 = 0$

(d)

$2 \log_2 xy - 3 \log_2 (xy) = 4$
 $\log_2 (xy)^2 - \log_2 (xy)^3 = 4 \log_2 2$

(1)

$\frac{x^2 y^2}{(xy)^3} = 2^4$

(1)

i.e. $x^2 y^2 = 16 (xy)^3$

(e) $L + P = \frac{5}{4}$ $L^2 + P^2 = (L+P)^2 - 2LP$

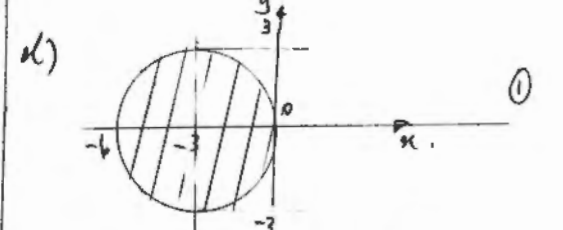
(1)

$= \left(\frac{5}{4}\right)^2 - 2 \times \frac{1}{4}$

$= \frac{25}{16} - \frac{2}{4}$

$= \frac{-3}{16}$

(1)



(e) Construction: Join AC.
 Proof: In $\triangle AXC$ & $\triangle AYC$

AC is common
 $CX = CY$ (Data)

$\angle ACX = \angle ACY$ (diagonal AC bisects angle of square)

$\therefore \triangle AXC \equiv \triangle AYC$ (SAS)

$\therefore \angle AXC = \angle AYC$ (corresponding angles of congruent triangles are equal)

(1)

(1)

(1)

(a) $k=0$

(1)

(b) $\log_4 128 = \frac{\log_2 128}{\log_2 4}$
 $= \frac{7}{2}$

(1)

(1)

(c) $\frac{x^3 - 27}{x^2 - 9} = \frac{(x-3)(x^2 + 3x + 9)}{(x-3)(x+3)}$

(1)

(1)

$= \frac{x^2 + 3x + 9}{x+3}$ $x \neq -3$

(d) $\frac{3x-5}{2} - \frac{9x-1}{5} < -6$

$5(3x-5) - 2(9x-1) < -60$

$15x - 25 - 18x + 2 < -60$

$-3x - 23 < -60$

$-3x < -37$

$x > 12\frac{1}{3}$

(e) $\frac{\log_7 8}{\log_7 2} = \log_2 8$

(1)

(1)

(f) $\left(\frac{4\sqrt{2}-3}{5+\sqrt{2}}\right) \left(\frac{5-\sqrt{2}}{5-\sqrt{2}}\right)$

$= \frac{20\sqrt{2} - 8 - 15 + 3\sqrt{2}}{25 - 2}$

$= \frac{23\sqrt{2} - 23}{23}$

$= \sqrt{2} - 1$

(2)

(g) $\sqrt{(x+1)^2} = |x+1|$

(1)

3^c) $P(x) = x^3 - 6x^2 + 11x - 6$
 $P'(x) = 3x^2 - 12x + 11$
 $P''(x) = 6x - 12$

for possible inflexion point

$P''(x) = 0$
 $6x - 12 = 0$
 $x = 2$

Since y, y', y'' etc for all x
 that y'' for change in concavity

$y''(1.9) = 6 \times 1.9 - 12$
 $= -0.6$

$y''(2.1) = 6 \times 2.1 - 12$
 $= 0.6$

> 0

∴ change in concavity

∴ Inflexion point, $x = 2$
 $y = 0$

$\Rightarrow (2, 0)$

4(a) Midpoint AC $\equiv \left(\frac{-14+8}{2}, \frac{18+12}{2} \right)$

$\equiv (-3, 15)$

Midpoint BD $\equiv \left(\frac{-8+2}{2}, \frac{27+3}{2} \right)$

$\equiv (-3, 15)$

∴ midpoints equal.

∴ ABCD is parallelogram

(diagonals AC & BD bisect each other)

(b) $2 \sin x + 3 \cos x$
 $\tan x = -\frac{3}{2}$
 $x = -56^\circ 19'$

$A = 150 - 32y + 2y^2$ $y = 5 + 4x - x^2$

(c) (i) $\frac{dA}{dx} = \frac{dA}{dy} \cdot \frac{dy}{dx}$
 $= (-32 + 4y)(4 - 2x)$

let $x = 0.5$ $y = 5 + 4 \times 0.5 - 0.25$
 $= 6.75$

∴ $\frac{dA}{dx} = (-32 + 4 \times 6.75)(4 - 2 \times 0.5)$
 $= -5 \times 3$
 $= -15$

(ii) At $x = 1.5$ $\frac{dA}{dx} = (-32 + 4y)(4 - 2x)$
 $y = 8.75$ $= 3$

At $x = 2.5$ $\frac{dA}{dx} = (-32 + 4y)(4 - 2x)$
 $y = 8.75$ $= -3$

(iii) For max/min values $\frac{dA}{dx} = 0$

$\therefore -32 + 4y = 0$ And $4 - 2x = 0$
 $y = 8$ $x = 2$

$\therefore 8 = 5 + 4x - x^2$

$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

$x = 1$ or 3 or 2 .

Test turning points using gradient
 $\frac{d^2A}{dx^2}$ where A, y are continuous {x: 0 to 4}

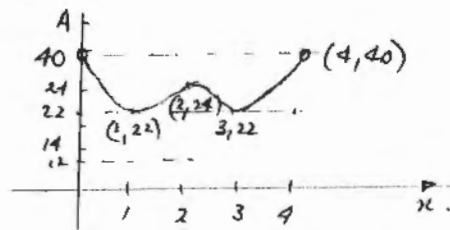
x	0.5	1	1.5	2	2.5	3	3.5
$\frac{dA}{dx}$	-15	0	3	0	-3	0	15

Since change in gradient from -ve to +ve at $x = 1$
 then $x = 1$ is a relative minimum

Since change in gradient from +ve to -ve at $x = 2$
 then $x = 2$ is a relative maximum

Since change in gradient from -ve to +ve at $x = 3$
 then $x = 3$ is a relative minimum

(iv)



(1) endpoints
 (1) Graph.